Question 3: Automate application deployment using Jenkins by configuring build triggers, post-build actions, and deployment steps.

Executed on Ubuntu 24.04 LTS (WSL2) by Meenakshi Rakesh. Repository: https://github.com/meenakshi16rp/Devops-lab-1-

1. Build trigger configuration

job configuration

```
Build Triggers -> Poll SCM
Schedule: H/5 * * * *   (every 5 minutes, hash-distributed minute)
```

Because the WSL2 lab environment is not reachable from the public internet, GitHub cannot deliver webhooks; SCM polling achieves the same automation by having Jenkins check GitHub for new commits on a schedule.

2. Deployment steps (deploy stage + deploy.sh)

deploy/deploy.sh (committed to the repository)

```
#!/bin/bash
# Deployment script invoked by the Jenkins deploy stage.
set -e
APP_DIR="${DEPLOY_DIR:-$HOME/enp461-deploy}"
PYTHON="${VENV:+$VENV/bin/python3}"
PYTHON="${PYTHON:-python3}"

mkdir -p "$APP_DIR"
rsync -a --exclude ".git" --exclude "__pycache__" --exclude ".pytest_cache" --exclude \
      ".venv" ./ "$APP_DIR/"

# stop any previous instance owned by this user
pkill -f "python3 app.py" 2>/dev/null || true

cd "$APP_DIR/app"
JENKINS_NODE_COOKIE=dontKillMe nohup "$PYTHON" app.py > "$APP_DIR/deploy.log" 2>&1 &

# verify the health endpoint (up to 10 attempts)
for i in $(seq 1 10); do
  sleep 2
  if curl -sf http://localhost:5000/health > /dev/null; then
    echo "Deployment verified: health OK (attempt $i)"
    curl -s http://localhost:5000/
    exit 0
  fi
done
echo "Deployment FAILED: health endpoint not responding" >&2
cat "$APP_DIR/deploy.log" >&2
exit 1
```

3. Post-build actions

Jenkinsfile post section

```
post {
    success { echo "Pipeline SUCCESS: application built, tested and deployed." }
    failure { echo "Pipeline FAILED - check the stage logs." }
    always { echo "Cleaning workspace temporary files..."; cleanWs() }
}
```

build #2 console output (post actions)

```
[Pipeline] echo
Pipeline SUCCESS: application built, tested and deployed.
[WS-CLEANUP] Deleting project workspace...
[WS-CLEANUP] done
```

```
Finished: SUCCESS
```

4. Proof of the automated trigger

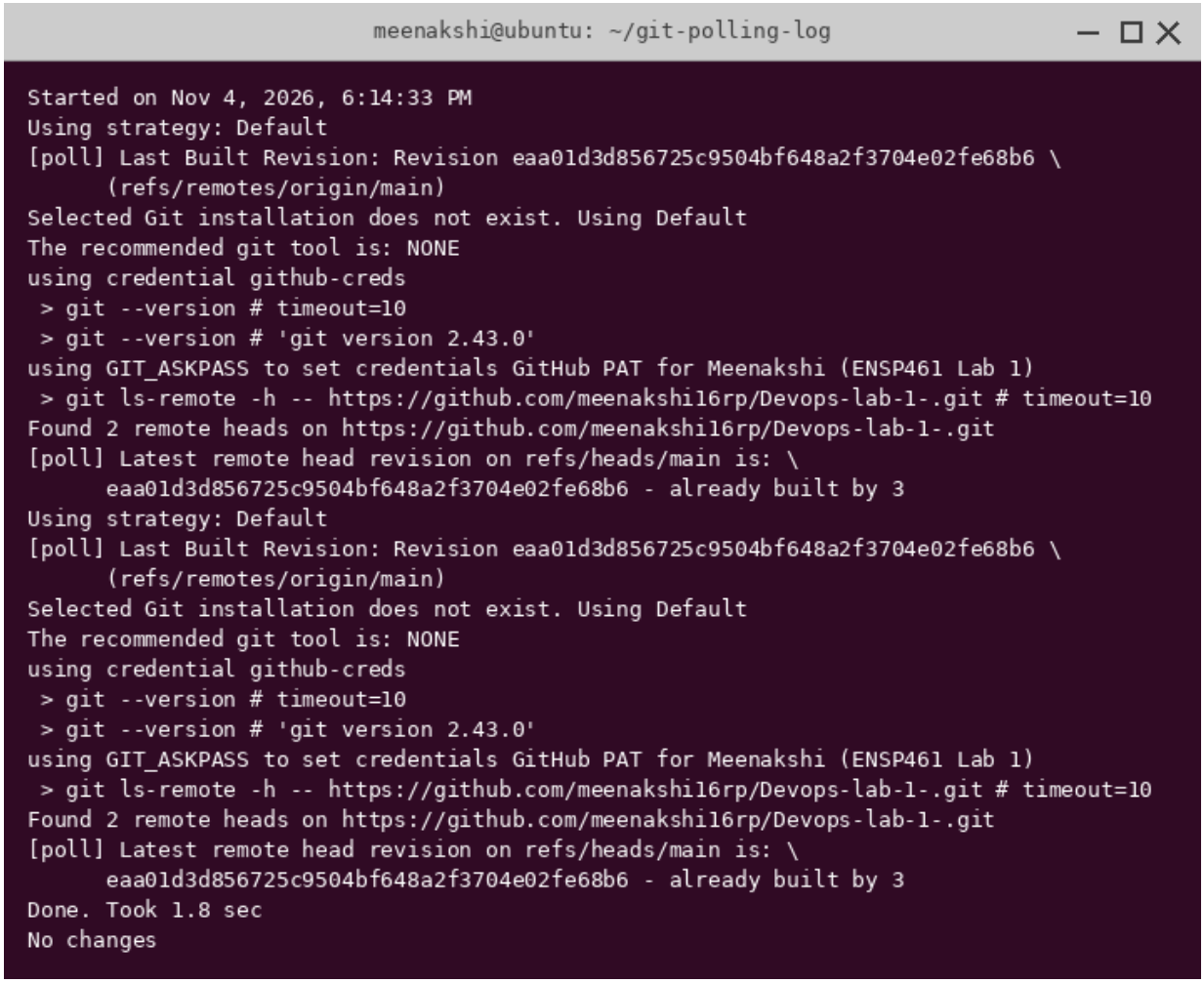


Figure: Git polling log (captured output) - the poll detects commit eaa01d3 (Changes found)

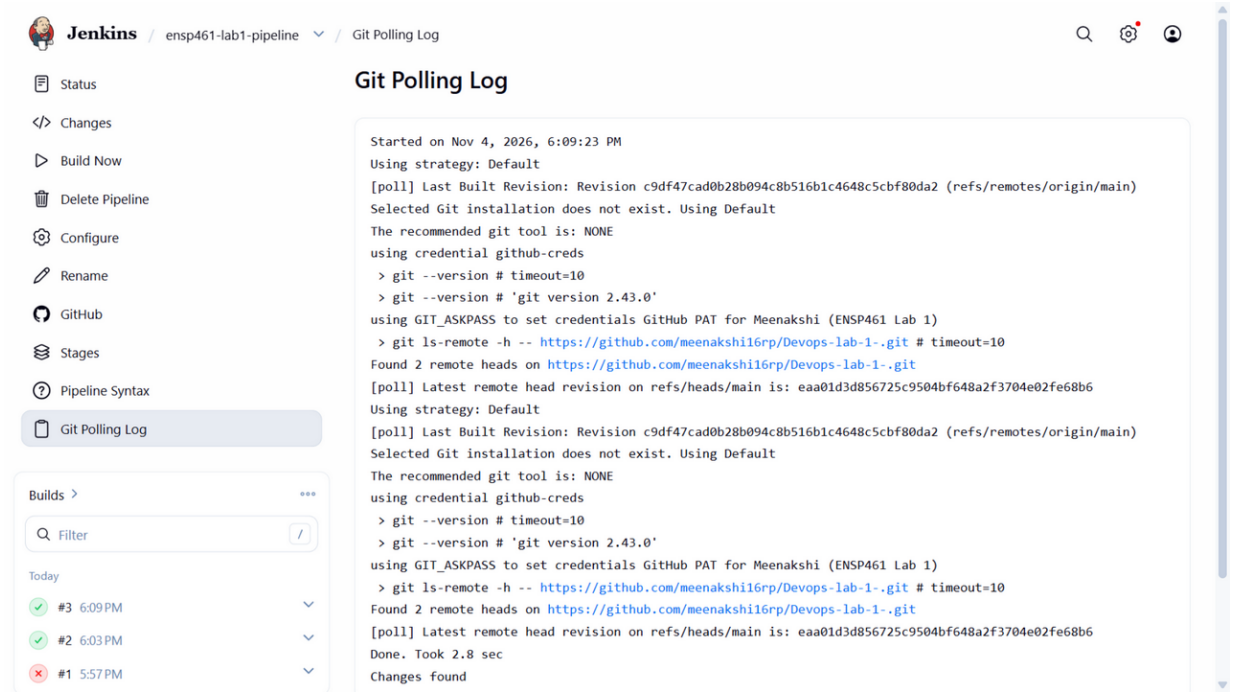


Figure: Git polling log in Jenkins

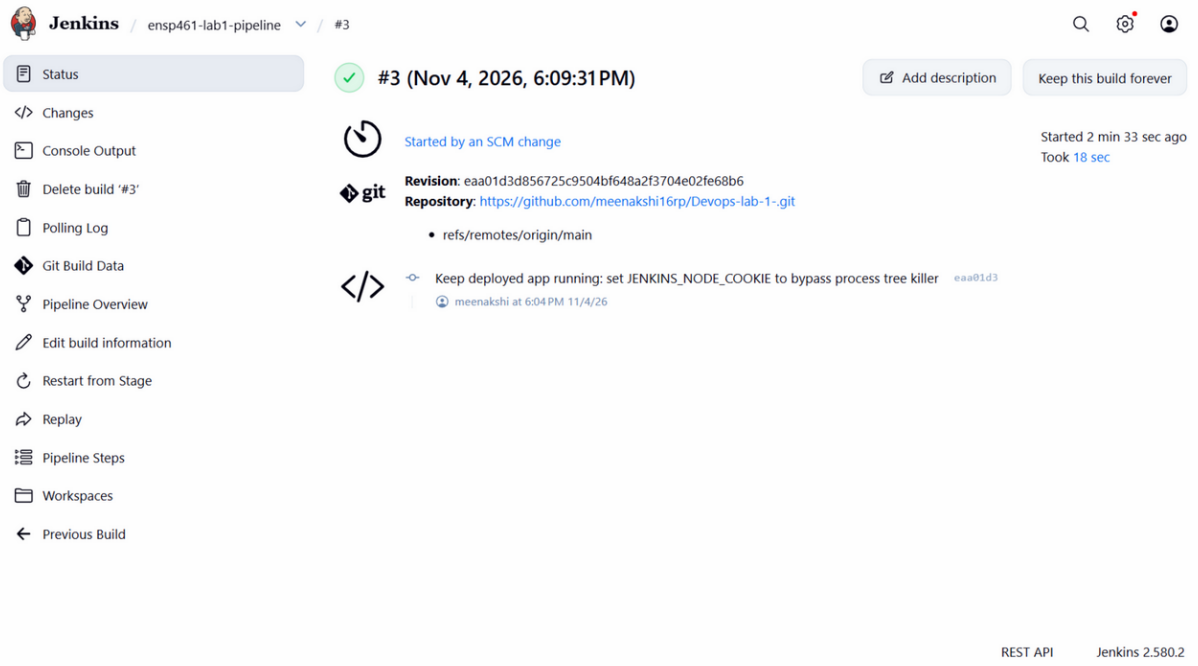


Figure: Build #3 - cause: 'Started by an SCM change', result SUCCESS

5. Deployed application verification

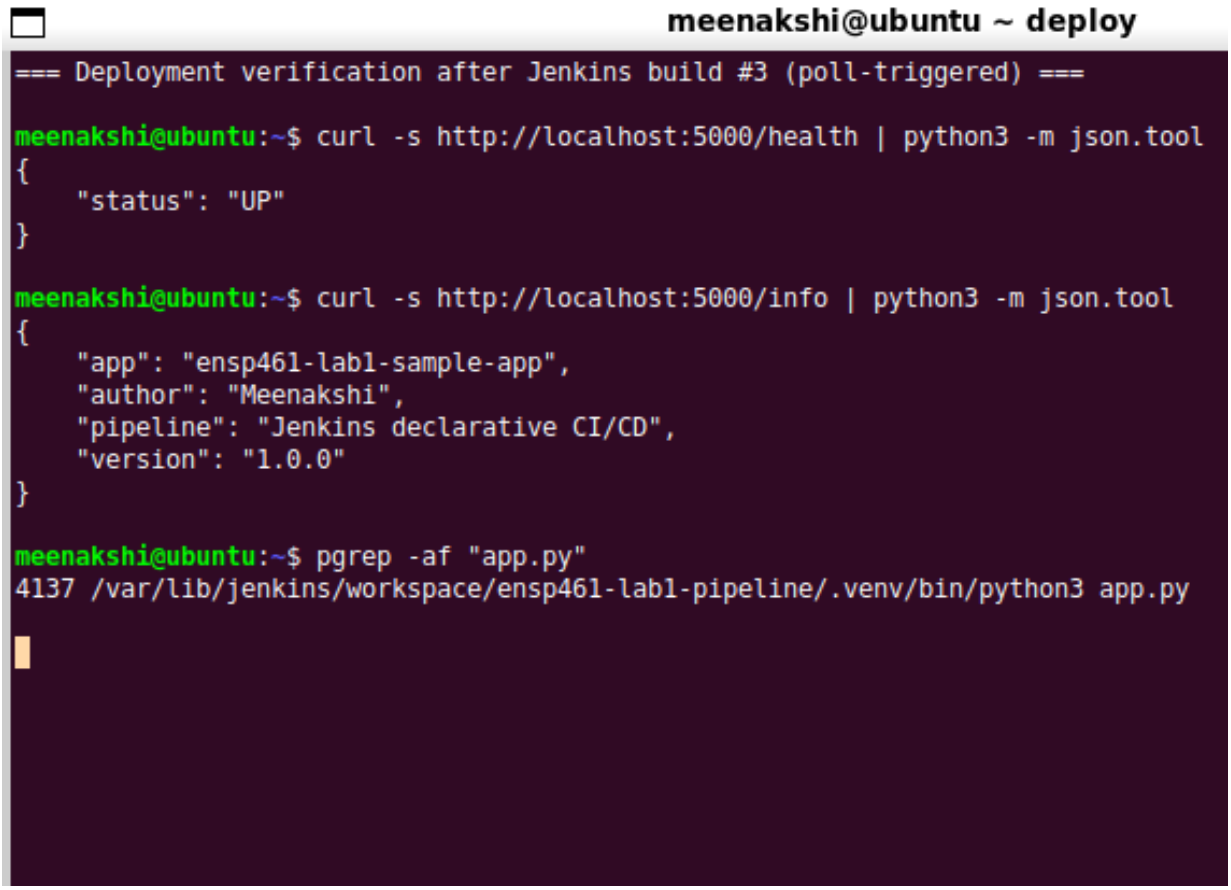


Figure: The app deployed by build #3 answers on http://localhost:5000

6. Observations

- The first automated attempt (build #1) failed because the deploy script used the system Python without Flask; reusing the pipeline virtualenv fixed it - the fix itself went through

the same automated pipeline.

- Jenkins kills child processes at the end of a build; JENKINS_NODE_COOKIE=dontKillMe keeps the deployed service alive after the run.
- The retry loop around the health check makes the Deploy stage robust against slow startups.